

One Record per Click: Distilling a Matured-Label Teacher for Delayed-Feedback Conversion Prediction

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Abstract. Conversion-rate models are trained on a stream in which a click’s label is not yet known. Many production pipelines resolve this by waiting a fixed time Δ , writing the label $\mathbf{1}[\text{conversion} \leq \text{click} + \Delta]$, and never revisiting that example. We show that most published delayed-feedback methods quietly assume the opposite. Instrumenting fourteen methods and counting the training records each emits per click over a full replay of Criteo-DFD, we find that nine exceed one, by factors from 1.08 to 3.00 — and that the duplicating VANILLA baseline against which the field reports relative improvement is itself inadmissible. The constraint collapses the design space to two label pipelines, a timely biased one and a matured unbiased one, and every admissible method combines them. We introduce DISTIL, which spends the matured pipeline on one offline teacher rather than a small online auxiliary and hands its soft label to a single-visit student. Across a five-rung serving-capacity ladder DISTIL is the strongest admissible method, beating the best prior one by up to +0.0025 AUC ($p < 10^{-5}$) at the rung where student and teacher have *equal* capacity, so the gain is attributable to matured labels rather than to capacity transfer. We also find the non-causal Oracle is not an upper bound for small students: below 10^5 parameters a soft-labelled admissible method overtakes it.

Keywords: Conversion rate prediction · Delayed feedback · Knowledge distillation · Online advertising.

1 Introduction

Conversion-rate (CVR) prediction decides what an advertising system shows and what it charges, and it is trained continuously on a stream. Its central difficulty is that the label arrives late: a click is observed immediately, but the conversion it may cause can follow minutes, days or weeks afterwards [1]. A model trained on the labels observed so far is therefore trained on a censored, downward-biased target.

The literature treats this as an estimation problem, and a decade of work has produced importance weighting [14, 13], label correction [2, 11], duplicated replay streams [6, 5, 3], multi-window ensembles [7, 9, 15] and influence-function retrofits [4]. Almost all share an unstated systems assumption: that a click may be written into the training stream more than once — re-emitted when its conversion lands, replayed when its window closes, or relabelled at several horizons.

Many production pipelines cannot do this. A common contract is:

A click produces *exactly one* training example. It is emitted after a fixed wait Δ , carries the label $\mathbf{1}[\text{conv_ts} \leq \text{click_ts} + \Delta]$, and is never revisited. If the conversion has not arrived by then, the example is negative permanently.

The reasons are mundane and therefore durable: amending a materialised example later means mutable training data, duplicate-key handling, and a reproducibility story that breaks on every backfill, whereas freezing the label at Δ keeps the dataset append-only and the pipeline replayable. We take this as a hard constraint and ask what survives it.

Contributions.

1. **A measurement, not an assertion.** We instrument the feature path of fourteen methods and count training records per click over a full replay (Sect. 5.2). Nine exceed one. Two measurement traps — traffic variation and window length — both *understate* violations; we control for both.
2. **A reformulation.** The constraint admits exactly two label pipelines, and every compliant method combines them (Sect. 3). It also shows the field’s VANILLA anchor is inadmissible, so relative improvements reported against it are not production-relevant quantities.
3. **DISTIL**, which spends the matured pipeline on one offline teacher instead of a small online auxiliary (Sect. 4). It is the strongest admissible method we measure, and the student is compliant by construction — verified by the same instrument.
4. **A capacity study** over five serving-scale rungs (Sect. 5.4), the axis a latency-bound ranker is actually constrained on and one the delayed-feedback literature does not report. It yields the paper’s most surprising result: for small students the non-causal Oracle is not an upper bound.

2 Related Work

We organise prior work by whether it can be expressed as one record per click, because that is what the constraint decides. Sect. 5.2 gives measured counts; here we give the mechanism.

Duplication and replay. The fake-negative family writes the click as a negative and re-emits it as a positive when the conversion arrives [6]. ES-DFM samples an elapsed-time cut-off and duplicates the delayed positive [13]; DEFER additionally replays real negatives once their window closes [5]; DDFM runs both streams together [3]; DEFUSE adds a label correction on top of the duplicated stream [2]. Duplication *is* the mechanism, so the second record is not an implementation detail that can be removed.

Multi-window ensembles and retrofits. FTP trains K maturity-gated tasks and a policy over them [7]; MISS duplicates each delayed positive into every head’s stream [9]; PI interpolates two fixed-window predictors [15]. Each head relabels the same click at its own horizon, so the record count scales with the number of heads. IF-DFM instead applies an influence-function update when a label reverses [4]; revisiting the example is the method, not a side effect.

Single-record methods. A smaller group reweights or relabels the one fresh record instead of adding another. FSIW corrects the feedback shift by importance weighting [14]; ULC replaces the observed label with a corrected soft label [11]; TWICE keeps its CVR tower on a single fresh record and routes conversion-clock events to a separate delay head [8]. These are what the constraint leaves standing. ULC is the comparator that matters for us: it has the same functional form as our method and differs only in where its soft label comes from.

Distillation with privileged information. Training-time-only signals have been distilled into serving models before, notably privileged features at Taobao [12], and soft labels are known to carry lower target variance than a hard Bernoulli label [10]. Our teacher’s privilege is neither a feature nor scale but *time*: it is allowed to wait.

3 Problem Formulation

3.1 Two clocks, one wait

For click i let x_i be its features and C_i its click time. If a conversion follows, let V_i be its arrival time and $D_i = V_i - C_i$ the delay; otherwise $D_i = \infty$. Following the long-horizon formulation of [8], a business *attribution window* v fixes the served target $Y_i^{(v)} = \mathbf{1}[D_i \leq v]$, clicks are observed on a *click clock*, and conversions arrive on a separate *conversion clock*. Training must predict $Y^{(v)}$ from a record written before $Y^{(v)}$ is known.

3.2 The single-visit constraint

Let Δ be the pipeline’s wait. The contract of Sect. 1 imposes two obligations:

R1 (single visit) click i contributes exactly one training record, ever;

R2 (frozen label) that record’s label is $\mathbf{1}[D_i \leq \Delta]$ for the Δ at which it was emitted; no later-arriving conversion may add or change supervision.

A method may fail either independently: duplication fails R1, and reading $Y^{(v)}$ on a record released before v fails R2.

3.3 Two admissible pipelines

Because a record is written once and Δ is the only free quantity, the constraint admits exactly two useful pipelines over the same clicks:

$$D_{\text{fresh}} = \{(x_i, Y_i^{(o)})\}, \quad \Delta = o \ll v \quad \text{timely, biased low,} \quad (1)$$

$$D_{\text{mat}} = \{(x_i, Y_i^{(v)})\}, \quad \Delta = v \quad \text{unbiased, stale by } v. \quad (2)$$

For D_{fresh} , $\mathbb{E}[Y^{(o)} | x] = p_v(x) F(o | x) < p_v(x)$ where F is the conditional delay CDF, so it is biased low by exactly the fraction of converters not yet observed; on Criteo-DFD at $v = 7$ d that factor is 0.47, i.e. a model trained on D_{fresh} alone under-predicts by more than half. D_{mat} carries the true target but its most recent row is v old.

R1 forbids training the served model on both. What remains is to use D_{mat} to *modify the target* of the single fresh record — which is what ULC, FSIW and TWICE all do. We differ only in what the matured pipeline is spent on.

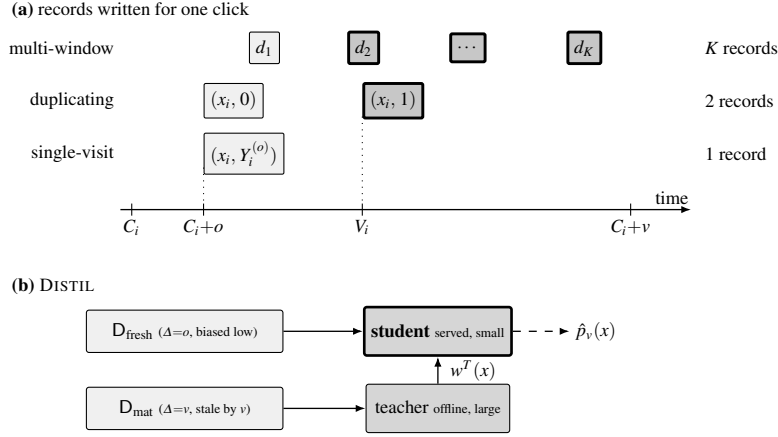


Fig. 1. (a) Records written for a single click. A compliant method writes one; duplicating and multi-window methods write more, and thick-bordered records are the second and later visits that violate R1. (b) DISTIL spends the matured pipeline on one offline teacher, whose soft label retargets the single fresh record, so the served student remains single-visit

4 Method

4.1 An offline teacher on the matured pipeline

The teacher trains only on clicks whose window has closed, $C_i + v \leq T_n$ at cutoff T_n , under the true label $Y^{(v)}$. It is causal by construction — every training row’s outcome was knowable at T_n — and is itself R1/R2-compliant at $\Delta = v$. It carries two read-outs on one trunk:

$$p_v^T(x) = P(Y^{(v)} = 1 \mid x), \quad w^T(x) = P(Y^{(v)} = 1 \mid Y^{(o)} = 0, x). \quad (3)$$

The second is the estimand the field shares — it is ULC’s w , ES-DFM’s delayed-positive rate and DDFM’s z_1 — so comparing against ULC isolates the estimator and nothing else. The teacher is *training-only* and therefore unbound by the serving budget: that asymmetry is the whole method, and what it buys is the right to wait.

4.2 The student

The student writes one record per click, at age o , with target

$$\tilde{y}_i = Y_i^{(o)} + (1 - Y_i^{(o)}) w^T(x_i), \quad (4)$$

minimising binary cross-entropy against it. Equation (4) is exactly ULC’s functional form: an observed positive stays positive, an observed negative is softened by the probability it converts later. The student never sees a second record and never sees a label that arrived after its own release, so it satisfies R1 and R2 at the same level as ULC and FSIW — verified with the instrument of Sect. 5.2, not by inspection.

Two implementation points matter and are easy to get wrong. The teacher’s learning rate is fixed and *decoupled* from the student’s: sharing it makes the per-arm sweep tune two models with one scalar, which cost 0.0065 AUC when it happened to FSIW’s auxiliary in our harness. And the teacher’s soft label is withheld until it has seen a minimum number of matured rows, because a barely-trained head asserting 0.5 poisons a one-pass run permanently.

5 Experiments

5.1 Setup

Data and protocol. Criteo-DFD [1], 60 days of clicks with conversion timestamps. We follow the streaming replay of [8]: pre-train on days $[0, 30)$, then stream $[30, 60)$ with hourly cut-offs; at each cut-off a model consumes the records released in that hour and then predicts every click of the next hour exactly once, scored retrospectively against $Y^{(v)}$. Base observation age $o = 1$ h, minibatch 8192, AdamW, and a learning rate selected *per arm and per capacity* from $\{10^{-6}, 2.5 \times 10^{-6}, 10^{-5}, 5 \times 10^{-5}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ on a held-out chronological slice of the pre-training period. We report $v = 7$ d, five seeds, and paired t -tests over matched seeds (* $p < 0.05$, ** $p < 0.01$).

Student ladder. The served model must be small, so capacity is swept over five rungs S0–S4 varying the categorical hash-table size ($36 \rightarrow 131,072$ rows, $26\text{ K} \rightarrow 9.6\text{ M}$ parameters) at a fixed embedding dimension of 8. The teacher is held at 9.5 M parameters throughout, so any trend across rungs is attributable to the student. At S4 student and teacher are within 1.3% of the same parameter count, which is the cleanest available control for capacity transfer.

Arms. ORACLE labels the fresh record with $Y^{(v)}$; it is not causal and exists only as a reference. FRESH trains on D_{fresh} uncorrected, WAIT on D_{mat} . REWEIGHT is FSIW, CORRECT is ULC, and we include TWICE. DISTIL-NOKD is the mandatory own-stream ablation: the identical pipeline with the teacher trained but its label unused.

5.2 RQ1: which methods can be run at all?

Table 1 reports the measurement. We instrument the feature path of every arm and count training records over a full 720-hour replay, charging only gradient-carrying forwards so that merely *scoring* a historical click is not counted. Two traps had to be controlled, and ignoring either understates violations.¹

Nine of fourteen methods exceed one record per click. Two counts are exact integers and confirm the instrument: PI emits 2.00, one per fixed-window head, and FTP 3.00,

¹ *Traffic*: dividing records consumed in an hour by clicks released in that hour conflates duplication with Criteo’s diurnal variation, since a maturity-gated stream reads clicks from v ago when the rate differed — that way PI scores a spurious 12.2, purely because its 14-day head reads an hour carrying $\sim 6\times$ the traffic of the denominator hour. We divide by *distinct clicks touched*. *Window length*: over 48 h PI measures 1.00 with a maximum of 1, because its two visits are 10.5 days apart and the second falls outside the window. We run the full stream and score only clicks whose entire revisit horizon fits inside it.

Table 1. Training records per click, measured over a full 720-hour replay at $v = 7$ d. Counts are per *distinct click touched*, so traffic variation cannot inflate them, and the window covers every method’s longest revisit lag. [†] TWICE’s served tower is single-visit; its delay head reads the conversion clock

method	records/click	max	R1	R2
<i>admissible</i>				
Fresh	1.00	1	✓	✓
Wait / teacher	1.00	1	✓	✓
FSIW	1.00	1	✓	×
ULC	1.00	1	✓	×
Twice	1.00	1	✓	✓ [†]
<i>inadmissible</i>				
Vanilla	1.08	2	×	×
DFM	1.34	8	×	×
ES-DFM	1.08	2	×	×
DEFUSE	1.08	2	×	×
DDFM	1.89	2	×	×
PI	2.00	2	×	×
MISS	2.05	3	×	×
IF-DFM	2.90	99	×	×
FTP	3.00	3	×	×

one per task plus its prophet. IF-DFM touches one click up to 99 times. Only FSIW, ULC and TWICE keep the served model single-visit.

The field’s anchor is inadmissible. VANILLA — “each click trained once at base observation age o ” — measures 1.08 records per click, because matching any published VANILLA row requires also emitting the late conversion as a positive.² Relative improvement reported against it is therefore normalised by a baseline a single-visit pipeline cannot run.

On auxiliaries. FSIW, ULC and TWICE emit one record for the served model but fit a *separate* model on matured data, so they satisfy R1 and sit at the boundary of R2. We read the constraint as governing the served model’s training pipeline, under which they comply; the alternative reading — that no later-arriving conversion may inform anything — leaves only Waiting(Δ) standing and declares the problem unsolvable rather than selecting a method. We adopt the former and say so, because the choice determines which methods are admissible. DISTIL is built to sit at exactly this level and no lower.

5.3 RQ2: performance under the constraint

Table 2 gives the main result. DISTIL is the strongest admissible method at every rung but S0, where it ties CORRECT (-0.0003 , $p = 0.59$); elsewhere it leads by $+0.0008$

² The fresh record alone predicts 0.107 against a 0.227 base rate and cannot reproduce any published VANILLA log-loss; the reading that does is fresh label plus a positive for conversions arriving after o .

Table 2. AUC across the serving-capacity ladder, $v = 7$ d, five seeds. Significance on DISTIL is against CORRECT, the strongest prior admissible method, paired over matched seeds

student parameters	S0 26K	S1 46K	S2 115K	S3 645K	S4 9.6M
<i>admissible: one record per click</i>					
Fresh = Distil-NoKD	0.8077	0.8127	0.8182	0.8227	0.8218
Wait	0.8105	0.8156	0.8212	0.8286	0.8321
Reweight	0.8218	0.8269	0.8334	0.8388	0.8396
Twice	0.8120	0.8217	0.8260	0.8305	0.8327
Correct	0.8276	0.8321	0.8372	0.8411	0.8411
Distil	0.8273	0.8328*	0.8377	0.8416**	0.8437**
<i>inadmissible: > 1 record per click</i>					
Vanilla	0.8222	0.8280	0.8347	0.8403	0.8412
Oracle (<i>not causal</i>)	0.8256	0.8311	0.8382	0.8438	0.8465

(S1, $p = 0.018$), $+0.0004$ (S2, $p = 0.089$), $+0.0005$ (S3, $p = 0.005$) and $+0.0025$ (S4, $p = 3 \times 10^{-6}$).

The ordering of the admissible field is stable across the ladder and worth stating, because it is not the ordering the literature reports: CORRECT > REWEIGHT > TWICE > WAIT > FRESH. TWICE, the most recent method and the strongest overall in its own evaluation, places fourth here — not because the method fails but because its CVR tower is deliberately fresh-only, so it starts from the FRESH floor and must climb back with its delay head alone.

The gain is not capacity transfer. DISTIL-NOKD — the same pipeline with the teacher trained but its label discarded — lands exactly on FRESH at every rung, as it must, since the two are the same algorithm. The distillation gain over that floor is $+0.0189$ to $+0.0218$ AUC, significant at $p < 10^{-6}$ throughout. Crucially, at S4 the student and teacher hold the same parameter count to within 1.3%, and DISTIL still leads CORRECT by its largest margin. The advantage therefore cannot be capacity flowing from a larger model into a smaller one; it is the matured label.

5.4 RQ3: the effect of student capacity

Fig. 2 reports the capacity sweep; three observations follow. The secondary metrics agree with AUC at the capacity-matched rung S4, where DISTIL reaches PR-AUC 0.5920 and LL 0.3496 against CORRECT’s 0.5875 and 0.3513, with the non-causal Oracle at 0.5953 and 0.3468.

The delayed-feedback penalty grows with capacity. The recoverable gap Oracle — FRESH widens from 0.0179 at S0 to 0.0247 at S4. Small models are intrinsically less hurt by label censoring, because they cannot exploit the resolution the missing labels would have supplied.

The Oracle is not an upper bound for small students. CORRECT — Oracle is $+0.0019$ at S0 ($p = 0.006$), $+0.0010$ at S1 ($p = 0.10$), then turns negative: -0.0010 (S2), -0.0026

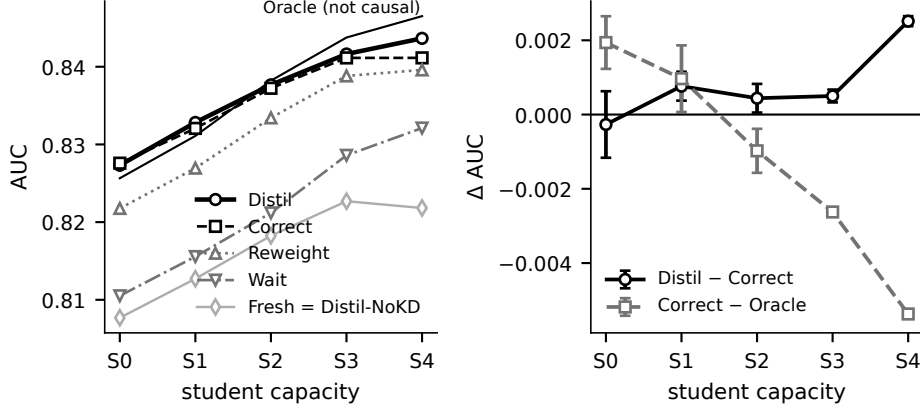


Fig. 2. Left: AUC against serving capacity; all admissible methods plus the non-causal Oracle. Right: two margins with 95% intervals over five paired seeds. CORRECT – Oracle crosses zero between S1 and S2 — below that, a soft-labelled admissible method beats the Oracle

Table 3. Teacher design, student fixed at S2. Δ is against the default teacher

variant	AUC Δ vs default
DISTIL, teacher 9.5 M (default)	0.8377 –
teacher 57 M	0.8360 -0.0016**
teacher 60 M	0.8361 -0.0016**
teacher trains only w	0.8376 -0.0000
teacher label unused (NoKD)	0.8182 -0.0194**

(S3), -0.0054 (S4), all significant from S2 on. Below roughly 10^5 parameters a corrected soft label is *better supervision than the true label*. This is consistent with the variance account of distillation [10]: a soft target has lower variance than a Bernoulli draw, and variance reduction dominates bias when capacity is scarce. It also means an evaluation conducted only at large capacity — as the delayed-feedback literature does — reports a penalty that a deployable model does not fully pay.

The margin over prior work widens with capacity. DISTIL – CORRECT rises from a tie at S0 to $+0.0025$ at S4. The teacher’s soft label carries more information than the small online auxiliary’s, but only a student with enough capacity can absorb the difference; at S0, with 36 hash rows, both targets are equally unrepresentable.

5.5 RQ4: which part of the teacher matters?

Table 3 varies the teacher at a fixed student. Removing the soft label entirely costs -0.0194 , which is the whole method. Restricting the teacher to train only the head the student consumes changes nothing (-0.0000), so the second read-out is free. Enlarging the teacher from 9.5 M to 57 M or 60 M parameters *hurts* by -0.0016 — the subject of the next section.

5.6 Why the teacher’s margin is small on this corpus

Two comparisons are easily confused here, and separating them explains both why the teacher helps and why enlarging it does not.

Teacher versus student: a large margin that shrinks with student size. In NE at $v = 7$ d the teacher reaches 0.7405 against a student floor of 0.7821 at S0 and 0.7522 at S4 — headroom of 0.0416 and 0.0117 respectively, a factor of 3.6. Capacity matters at the small end, which is the end a latency-bound ranker occupies.

Teacher versus larger teacher: no margin at all. Scaling past ~ 9.4 M parameters does not help and then hurts: NE 0.7405, 0.7454 and 0.7489 at 9.4, 57 and 60 M. This is a property of the corpus. Criteo-DFD’s nine categorical fields hold 135,151 distinct values in total against a table provisioned at 131,072 rows *per field*, so a model at full resolution has already absorbed every input the data can distinguish and further parameters buy access to nothing. The saturation point and the teacher’s optimum coincide. One-pass streaming compresses the ladder further: a batch-retrained variant of the same ladder spans 0.0536 NE from S0 to S4 where ours spans 0.0300, and extra teacher epochs do not recover the difference (0.8381/0.8385/0.8345/0.8381 AUC at 1/2/4/8 epochs).

An expectation, not a result. These measurements bound teacher headroom on a corpus whose vocabulary is $\sim 10^5$ and whose saturation point is $\sim 10^7$ parameters. Production catalogues and user identifiers are orders of magnitude larger, so the saturation capacity is correspondingly higher while the student stays pinned by the serving budget; we therefore expect a substantially larger teacher–student gap, and margin for distillation, than the 0.0117–0.0416 NE here. *We do not measure this.* It is also why we report the gain against student capacity rather than at one point: the slope, not the level, is what transfers.

6 Limitations

One public corpus with a single conversion event, so multi-objective consolidation is motivation rather than experiment. The constraint is one production contract among several: systems that do support replay may use the methods we rule out, and for them our audit is a cost accounting rather than a prohibition. Whether R2 binds auxiliary models is a reading we adopt explicitly (Sect. 5.2), not a fact about the constraint. The ladder varies table rows at fixed embedding width; other capacity axes may differ. The extrapolation in Sect. 5.6 is an expectation grounded in a measured saturation mechanism, not a result.

7 Conclusion

Delayed-feedback methods are usually compared on accuracy alone. We added a systems constraint that many production pipelines already impose — one training record per click, label frozen at a fixed wait — and measured who satisfies it. Nine of fourteen published methods do not, including the baseline the field normalises against. Among

those that do, spending the matured-label pipeline on a single offline teacher and distilling it into a single-visit student is the strongest option we find, and at equal student–teacher capacity the gain is attributable to the teacher’s matured labels rather than to its size. The capacity sweep also shows that the delayed-feedback penalty, and the value of a corrected label, both depend strongly on the size of the served model — to the point that below 10^5 parameters a corrected soft label beats the true one. Evaluating at a single capacity, as the literature does, hides this.

References

1. Chapelle, O.: Modeling delayed feedback in display advertising. In: Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD ’14). pp. 1097–1105. Association for Computing Machinery, New York, NY, USA (2014). <https://doi.org/10.1145/2623330.2623634>
2. Chen, Y., Jin, J., Zhao, H., Wang, P., Liu, G., Xu, J., Zheng, B.: Asymptotically unbiased estimation for delayed feedback modeling via label correction. In: Proceedings of the ACM Web Conference 2022 (WWW ’22). pp. 369–379. Association for Computing Machinery, New York, NY, USA (2022). <https://doi.org/10.1145/3485447.3511965>
3. Dai, S., Zhou, Y., Xu, J., Wen, J.R.: Dually enhanced delayed feedback modeling for streaming conversion rate prediction. In: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM ’23). pp. 390–399. Association for Computing Machinery, New York, NY, USA (2023). <https://doi.org/10.1145/3583780.3614856>
4. Ding, C., Wu, J., Yuan, Y., Li, C., Wang, X., Wang, D., Yang, F., Rabinovich, A.: Delayed feedback modeling with influence functions. Proceedings of the AAAI Conference on Artificial Intelligence **40**(17), 14648–14656 (2026). <https://doi.org/10.1609/aaai.v40i17.38483>
5. Gu, S., Sheng, X.R., Fan, Y., Zhou, G., Zhu, X.: Real negatives matter: Continuous training with real negatives for delayed feedback modeling. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD ’21). pp. 2890–2898. Association for Computing Machinery, New York, NY, USA (2021). <https://doi.org/10.1145/3447548.3467086>
6. Ktena, S.I., Tejani, A., Theis, L., Myana, P.K., Dilipkumar, D., Huszar, F., Yoo, S., Shi, W.: Addressing delayed feedback for continuous training with neural networks in ctr prediction. In: Proceedings of the 13th ACM Conference on Recommender Systems (RecSys ’19). pp. 187–195. Association for Computing Machinery, New York, NY, USA (2019). <https://doi.org/10.1145/3298689.3347002>
7. Li, H., Pan, F., Ao, X., Yang, Z., Lu, M., Pan, J., Liu, D., Xiao, L., He, Q.: Follow the prophet: Accurate online conversion rate prediction in the face of delayed feedback. In: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR ’21). pp. 1915–1919. Association for Computing Machinery, New York, NY, USA (2021). <https://doi.org/10.1145/3404835.3463045>
8. Li, K., Wang, K., Wang, Z., Sha, T., Yan, M., Cheng, Y., Liu, X.: TWICE: Two-clock, two-window learning for long-horizon conversion prediction in online advertising (2026)
9. Liu, Q., Ao, X., Guo, Y., He, Q.: Online conversion rate prediction via multi-interval screening and synthesizing under delayed feedback. Proceedings of the AAAI Conference on Artificial Intelligence **38**(8), 8796–8804 (2024). <https://doi.org/10.1609/aaai.v38i8.28726>
10. Menon, A.K., Rawat, A.S., Reddi, S.J., Kim, S., Kumar, S.: A statistical perspective on distillation. In: Proceedings of the 38th International Conference on Machine Learning (ICML) (2021)

11. Wang, Y., Sun, P., Zhang, M., Jia, Q., Li, J., Ma, S.: Unbiased delayed feedback label correction for conversion rate prediction. In: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '23). pp. 2456–2466. Association for Computing Machinery, New York, NY, USA (2023). <https://doi.org/10.1145/3580305.3599536>
12. Xu, C., Li, Q., Ge, J., Gao, J., Yang, X., Pei, C., Sun, F., Wu, J., Sun, H., Ou, W.: Privileged features distillation at taobao recommendations. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '20) (2020)
13. Yang, J.Q., Li, X., Han, S., Zhuang, T., Zhan, D.C., Zeng, X., Tong, B.: Capturing delayed feedback in conversion rate prediction via elapsed-time sampling. Proceedings of the AAAI Conference on Artificial Intelligence **35**(5), 4582–4589 (2021). <https://doi.org/10.1609/aaai.v35i5.16587>
14. Yasui, S., Morishita, G., Fujita, K., Shibata, M.: A feedback shift correction in predicting conversion rates under delayed feedback. In: Proceedings of The Web Conference 2020 (WWW '20). pp. 2740–2746. Association for Computing Machinery, New York, NY, USA (2020). <https://doi.org/10.1145/3366423.3380032>
15. Zhang, X., Li, W., Li, R., Fu, Z., Tang, T., Zhang, Z., Chen, W.Y., Noorshams, N., Jasapara, N., Ding, X., Wen, E., Feng, X.: Personalized interpolation: Achieving efficient conversion estimation with flexible optimization windows. In: Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25). Association for Computing Machinery, New York, NY, USA (2025)